NOTICE OF CHANGE

MIL-STD-975L (NASA)
NOTICE 1
12 APRIL 1994

NASA STANDARD ELECTRICAL, ELECTRONIC
AND ELECTROMECHANICAL (EEE) PARTS LIST

TO ALL HOLDERS OF MIL-STD-975L

1. THE FOLLOWING PAGES OF MIL-STD-975L HAVE BEEN REVISED AND SUPERSEDE THE PAGES LISTED:

NEW PAGE	DATE	SUPERSEDED PAGE	DATE
1.45	12 APRIL 1994	1.45	31 JANUARY 1994
1.46	31 JANUARY 1994	1.46	REPRINTED WITHOUT CHANGE
1.49	31 JANUARY 1994	1.49	REPRINTED WITHOUT CHANGE
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MIL-STD-975L
NOTICE 1

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2. RETAIN THIS NOTICE AND INSERT BEFORE TABLE OF CONTENTS.

3. Holders of MIL-STD-975L will verify that page changes and additions indicated above have been entered. This notice page will be retained as a check sheet. This issuance, together with appended pages, is a separate publication. Each notice is to be retained by stocking points until the military standard is completely revised or canceled.

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Project 59GP-K139

AMSC N/A
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MIL-C-23269, CAPACITORS
Fixed, Glass Dielectric, Established Reliability

Part number explanation:

M23269 /XX -XXXX

M23269 - Identifies CYR fixed, glass dielectric, established reliability capacitors conforming to MIL-C-23269.

/XX - Identifies the appropriate military specification sheet that uniquely specifies the capacitor family.

-XXXX - Uniquely specifies the nominal capacitance value, capacitance tolerance, rated dc voltage, and failure rate level.

Part Number	Control Specification	Style (1)	Capacitance		Working Voltage (Vdc) at +125 °C	Maximum Dissipation Factor (%)	Temperature		Minimum Insulation Resistance (Gigohms) at +25°C	FRL
			Range (pF)	Tolerance (±)			Range (°C)	Coefficient (ppm/°C)		Grade 1 and 2
M23269/01-XXXX	MIL-C-23269/1	CYR10	0.5-300	0.25 pF 0.50 pF 1%, 2%, 5%	100	0.1	-55 to +125	140 ±25	100	S
M23269/02-XXXX	MIL-C-23269/2	CYR15	220-1,200	1%, 2%, 5%						
M23269/03-XXXX	MIL-C-23269/3	CYR20	560-3,300							
M23269/04-XXXX	MIL-C-23269/4	CYR30	3,600-6,200							

(1) Lead material and coating are specified in the detailed specification sheet for each device type.

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MIL-C-23269/1, STYLE CYR10 CAPACITORS
Fixed, Glass Dielectric, Established Reliability

Capacitance		Part Number M23269/01-
Value (pF)	Tolerance (±)	Grades 1 and 2 FRL = S
0.5	0.25pF	7001
1.0	0.25pF	7002
1.5	0.25pF	7003
2.2	0.25pF	7004
	0.50pF	7005
2.7	0.25pF	7006
3.0	0.25pF	7007
	0.50pF	7008
3.3	0.25pF	7009
3.6	0.25pF	7010
	0.50pF	7011
3.9	0.25pF	7012
4.3	0.25pF	7013
	0.50pF	7014
4.7	0.25pF	7015
5.1	0.25pF	7016
5.6	0.25pF	7017
	5%	7018
6.2	0.25pF	7019
	5%	7020
6.8	0.25pF	7021
	5%	7022
7.5	0.25pF	7023
	5%	7024
8.2	0.25pF	7025
	5%	7026
9.1	0.25pF	7027
	5%	7028
10	0.25pF	7029
	5%	7030
11	0.25pF	7031
	5%	7032
12	0.25pF	7033
	5%	7034
13	2%	7035
	5%	7036
15	2%	7037
	5%	7038
16	2%	7039
	5%	7040
18	2%	7041
	5%	7042
20	2%	7043
	5%	7044
22	2%	7045
	5%	7046
24	2%	7047
	5%	7048
27	1%	7049
	2%	7050
	5%	7051
30	1%	7052
	2%	7053
	5%	7054
33	1%	7055
	2%	7056
	5%	7057
36	1%	7058
	2%	7059
	5%	7060
39	1%	7061
	2%	7062
	5%	7063

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MIL-C-23269/3, STYLE CYR20 CAPACITORS
Fixed, Glass Dielectric, Established Reliability

Capacitance		Part Number M23269/03-
Value (pF)	Tolerance (±%)	Grades 1 and 2 FRL = S
560	1	7001
	2	7002
	5	7003
620	1	7004
	2	7005
	5	7006
680	1	7007
	2	7008
	5	7009
750	1	7010
	2	7011
	5	7012
820	1	7013
	2	7014
	5	7015
910	1	7016
	2	7017
	5	7018
1,000	1	7019
	2	7020
	5	7021
1,100	1	7022
	2	7023
	5	7024

Capacitance		Part Number M23269/03-
Value (pF)	Tolerance (±%)	Grades 1 and 2 FRL = S
1,200	1	7025
	2	7026
	5	7027
1,300	1	7028
	2	7029
	5	7030
1,500	1	7031
	2	7032
	5	7033
1,600	1	7034
	2	7035
	5	7036
1,800	1	7037
	2	7038
	5	7039
2,000	1	7040
	2	7041
	5	7042
2,200	1	7043
	2	7044
	5	7045
2,400	1	7046
	2	7047
	5	7048

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MIL-C-23269/3, STYLE CYR20 CAPACITORS
Fixed, Glass Dielectric, Established Reliability (Continued)

Capacitance		Part Number M23269/03-
Value (pF)	Tolerance (±%)	Grades 1 and 2 FRL = S
2,700	1	7049
	2	7050
	5	7051
3,000	1	7052
	2	7053
	5	7054
3,300	1	7055
	2	7056
	5	7057

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MIL-C-23269/4, STYLE CYR30 CAPACITORS
Fixed, Glass Dielectric, Established Reliability

Capacitance		Part Number M23269/04-
Value (pF)	Tolerance (±%)	Grades 1 and 2 FRL = S
3,600	1	7001
	2	7002
	5	7003
3,900	1	7004
	2	7005
	5	7006
4,300	1	7007
	2	7008
	5	7009
4,700	1	7010
	2	7011
	5	7012
5,100	1	7013
	2	7014
	5	7015
5,600	1	7016
	2	7017
	5	7018
6,200	1	7019
	2	7020
	5	7021

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MIL-C-39003, CAPACITORS
Fixed, Tantalum (Solid) Electrolytic, Polarized, Established Reliability

Part Number Explanation:

M39003 /XX -XXXX

M39003 - Identifies fixed, tantalum, electrolytic (solid electrolyte), established reliability capacitors that are hermetically sealed in metal cases.

/XX - Identifies the appropriate military specification sheet that uniquely specifies the capacitor family

-XXXX - Uniquely specifies the nominal capacitance value, capacitance tolerance, rated voltage, maximum dc leakage and dissipation factor, and failure rate level.

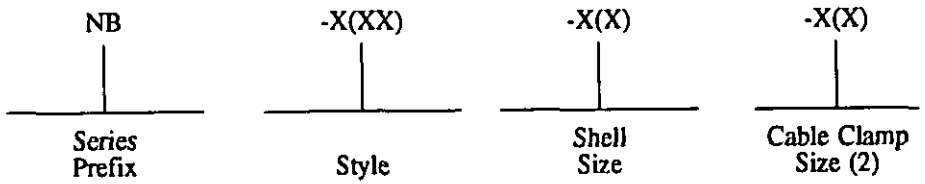
Part Number (1) (2)	Control Specification	Style	Capacitance		Rated Voltage (Vdc)	Operating Temperature Range (°C)	Configuration			FRL	
			Range (µF)	Tolerance (±%)			Case Type	Lead Type	Lead Material	Grade 1	Grade 2
M39003/10-XXXX* (3)	MIL-C-39003/10 Polarized	CSS13	0.12 - 220.0	10	10, 15, 20, 35, 50, 75	-55 to +125	Tubular	Axial	Tin-lead coated nickel; Solder coated nickel	C	B
M39003/01-XXXX (4)	MIL-C-39003/1 Polarized	CSR13	0.0047 - 4.7	10, 20	50				Tin-lead coated nickel; Solder coated nickel	C (5)	B
M39003/02-XXXX (4)	MIL-C-39003/2 Polarized	CSR09	0.047 - 15.0	10	10, 20, 35, 50, 75				Nickel-iron alloy	C (5)	B
M39003/10-XXXX* (3)	MIL-C-39003/10 Polarized	CSS33	1.2 - 560.0	10	10, 20, 35, 50				Tin-lead coated nickel; Solder coated nickel	C	B

(1) MIL-C-39003 capacitors shall not be used in power supply filters. Reference MIL-HDBK-978, Vol. 1, 2.6.7.2.
(2) Parts covered by this specification contain internal soldered connections which may reflow during installation. The A, A1, B, B1 case sizes are particularly susceptible and special precautions such as heat sinking are recommended when soldering onto boards.
(3) The symbol * completes the dash number thus: * = S for Sleeved, U for Unsleeved.
(4) CSR09 and CSR13 capacitors are sleeved (insulated case).
(5) All CSR13 and CSR09 capacitors must be subjected to the surge current test as specified by Appendix B. Reference MIL-HDBK-978, Vol.1, para. 2.6.7.2 and 2.6.7.3.

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40M39569 CONNECTORS
Electrical, Miniature, Circular, Environment Resisting, +200°C, Backshells

Part Number Explanation:



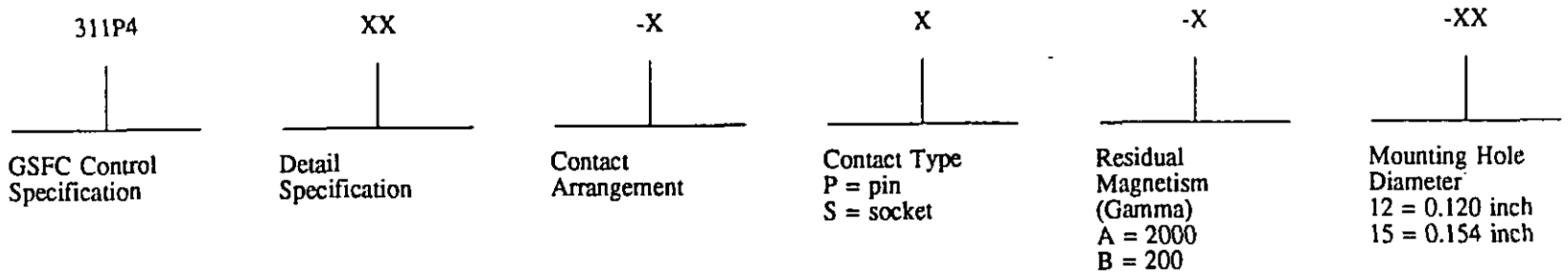
Part Number	Control Specification	Style	Shell Size
NB-C-X(X)	40M39569	Straight	8 through 24
NB-S-X(X)		Straight with strain relief	
NB-R-X(X)		90° with strain relief	
NB-SCT-X(X) (1)		Straight with cable tie	
NB-RCT-X(X) (1)		Straight with 90° cable tie	
NB-FCT-X(X)		Straight with 45° cable tie	
NB-RFI-X(X)-X(X)		Straight RFI	

(1) Add a "N" to the end of the part number to order without a ground lug.
(2) For RFI style only.

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GSFC S-311-P-4 CONNECTORS (1)
Connectors, Electrical, Subminiature, Rack and Panel Non-Magnetic

Part Number Explanation:



Part Number	Control Specification	Contact Type (3)	GSFC Contact Arrangement	Number of Contacts		Grades 1 and 2
				Standard Power	Coaxial and/or High Voltage	
311P407-XX-X-XX	GSFC S-311-P-4	Power, Size 22D	-1 -2 -3 -4 -5 -6	15 26 44 62 78 104	-	(2)

(1) These connectors may require material testing per NHB8060.1 for manned space flight applications.
(2) Parts may be used for Grade 1 and 2 applications.
(3) Contacts are supplied separately. See the following page listing S-311-P-4/06, /08 or /10 for appropriate part numbers.

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GSFC S-311-P-4 CONNECTORS (1)
Connectors, Electrical, Subminiature, Rack and Panel Non-Magnetic (Continued)

Part Number	Control Specification	Contact Type (3)	GSFC Contact Arrangement	Number of Contacts		Grades 1 and 2
				Standard Power	Coaxial and/or High Voltage	
311P405-XX-X-XX	GSFC S-311-P-4	Mixed, Power Size 20 & High Voltage or Coaxial Size 8 (4)	-6	4	1	(2)
			-7	0	3	
			-8	5	2	
			-9	10	1	
			-10	0	5	
			-11	5	4	
			-12	10	3	
			-13	15	2	
			-14	20	1	
			-15	0	8	
			-16	7	6	
			-17	12	5	
			-18	17	4	
			-19	17	4	
			-20	22	3	
			-21	25	2	
			-22	17	7	
			-23	32	4	
			-24	41	2	
			-25	46	1	
311P409-XX-X-XX		Power, Size 20	-1	9		
			-2	15		
			-3	25		
			-4	37		
			-5	50		

(1) These connectors may require material testing per NHB8060.1 for manned space flight applications.
(2) Parts may be used for Grade 1 and 2 applications.
(3) Contacts are supplied separately. See the following page listing S-311-P-4/06, /08 or /10 for appropriate part numbers.
(4) Coaxial contacts should be used for signals of 1 MHz frequency or less. The nylon hoods used on the high voltage contacts may not meet programs outgassing limits. See page 2.10 for contact part numbers.

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GSFC S-311-P-4 CONTACTS

Part Number Explanation:
For Coaxial and High Volume Contacts:

G	X	(X)	X	X
	C = Coaxial H = High Voltage	Blank = Straight R = Right Angle	P = Pin S = Socket	Wire Accommodation (See table below)

For Standard Power Contacts:

G	(X)	X	X
	Detail Specification	P = Pin S = Socket	Wire Accommodation (See table below)

Part Number	Control Specification	Size	Wire Accommodation
GC(X)X1	S-311-P-4/06	8	RG-178B/U
GH(X)X6 (1)	S-311-P-4/06	8	20 AWG, Max
G08X1	S-311-P-4/08	22D	22 to 28 AWG
G10X1	S-311-P-4/10	20	20 to 24 AWG

(1) The nylon hood used on high voltage contacts may not meet program outgassing limits.

GSFC S-311-P-10 CONNECTORS (1)

Connectors, Electrical, Rectangular, Miniature, Polarized Shell, Rack and Panel, Combination Solder, Coaxial and High Voltage Contacts, Non-Magnetic

Part Number Explanation:

311P10	(B)	-X(X)	X	-X	-XX
GSFC Control Specification	Socket Contact Designator No digit = contact springs B = split finger sleeved (PREFERRED)	Insert Arrangement	Contact Type P = pin S = socket	Residual Magnetism (Gamma) B = 200 C = 20	Mounting Hole Diameter 12 = 0.120 inch 15 = 0.154 inch

Part Number	Control Specification	Arrangement Number (2)	Number of Contacts		Grades 1 and 2
			Standard Power	Coaxial and or High Voltage (4)	
311P10-X(X)X-X-XX	GSFC S-311-P-10	1	9		(3)
		2	15		
		3	25		
		4	37		
		5	50		
		6	4	1	
		7	0	3	
		8	5	2	
		9	10	1	
		10	0	5	
		11	5	4	
		12	10	3	
		13	15	2	
		14	20	1	
		15	0	8	

(1) These connectors may require material testing per NHB8060.1 for manned space flight applications.
(2) Refer to Table II of the control specification for contact arrangements.
(3) Parts may be used for Grade 1 and 2 applications.
(4) Coaxial contacts should be used for signals of 1 Mhz frequency or less. The nylon hoods used on the high voltage contacts may not meet program outgassing limits. See page 2.10 for contact part numbers.

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GSFC S-311-P-10 CONNECTORS (1)
Connectors, Electrical, Rectangular, Miniature, Polarized Shell, Rack and Panel, Combination Solder, Coaxial and High Voltage Contacts, Non-Magnetic (Continued)

Part Number	Control Specification	Arrangement Number (2)	Number of Contacts		Grades 1 and 2
			Standard Power	Coaxial and or High Voltage (4)	
311P10-X(X)X-X-XX	GSFC S-311-P-10	16	7	6	(3)
		17	12	5	
		18	17	4	
		20	22	3	
		21	25	2	
		22	17	7	
		23	32	4	
		24	41	2	
		25	46	1	

(1) These connectors may require material testing per NHB8060.1 for manned space flight applications.
(2) Refer to Table II of the control specification for contact arrangements.
(3) Parts may be used for Grade 1 and 2 applications.
(4) Coaxial contacts should be used for signals of 1 Mhz frequency or less. The nylon hoods used on the high voltage contacts may not meet program outgassing limits. See page 2.10 for contact part numbers.

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SECTION 4: SUMMARY OF STANDARD DIODES

Page	Control Specification	Description	Type Designation	
			Grade 1	Grade 2 (1)
4.2	MIL-S-19500	Small signal	JANS	JANTXV
4.2		Power		
4.3		Multiple array		
4.3		Monolithic array		
4.4		Zener - voltage regulator		
4.5		Zener - voltage suppressor		
4.5		Bidirectional voltage suppressor		
4.6		FET - current regulator		
4.6		Schottky barrier		
4.7		Thyristor		

Refer to MIL-HDBK-978, Vol. 2, for construction and application information.

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

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MIL-S-19500 DIODES, SILICON

Part Number Explanation:

JAN	TXV	X (2)	IN	XXXX	B	-1
Military Qualified	Quality Level	Radiation Hardness Assurance (RHA)	Component Designation	Identification Number	Modifier	Metallurgical Bonding

	Total Ionizing Dose (RAD (Si))	Neutron Fluence (N/cm^2)
M	3×10^3	2×10^{12}
D	1×10^4	2×10^{12}
R	1×10^5	1×10^{13}
H	1×10^6	1×10^{13}

Description	Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Electrical Characteristics							Pkg Outline	Remarks
				V_F (Max) at I_F (pk)		I_R (Max) at V_R		I_o (Max avg)	C_t (Max) (pF)	t_{rr} (Max) (ns)		
				(V)	(A)	(μA)	(V)					
Small signal	1N645-1 (4)	1N645-1	/240	1.0	400 mA	50 nA	225	400 mA	20	—	A1	Rectifier
	1N647-1 (4)	1N647-1					400					
	1N649-1 (4)	1N649-1					600					
	1N6661	—	/587	1.0	400 mA	50 nA	225	500 mA	20	_	A1	Rectifier
	1N6662	—					400					
	1N6663	—					600					
	1N4148-1 (3)	1N4148-1 (3)	/116	1.0	10 mA	0.5	75	200 mA	4.0	5	DO-204	Fast switching
	1N4150-1	1N4150-1	/231	0.86	50 mA	0.1	50	200 mA	2.5	4		
		1N4153-1	/337	0.81	10 mA	0.05	50	150 mA	2.0	4		
		1N6638	/578	0.8	10 mA	0.5	125	300 mA	2.0	4.5		
	1N6642,U	1N6642,U	/578	1.0	10 mA	0.5	75	300 mA	5.0	5.0		
	1N6643,U	1N6643,U	/578	1.0	10 mA	0.5	50	300 mA	5.0	6.0		
Power	1N5415	1N5415	/411	1.5	9 A	1.0	50	3 A	550	150	A248	
	1N5416	1N5416					100		430	150		
	1N5417	1N5417					200		250	150		
	1N5418	1N5418					400		165	150		
	1N5419	1N5419					500		140	250		
	1N5420	1N5420					600		120	400		

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

(2) Include only for JANTXV and JANS product assurance level devices.

(3) New designs shall use MIL-S-19500/578.

(4) New designs shall use MIL-S-19500/587.

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MIL-S-19500 DIODES, SILICON (Continued)

Description	Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Electrical Characteristics							Pkg Outline	Remarks
				V_P (Max) at I_P (pk)		I_R (Max) at V_R		I_o (Max avg) (A)	C_t (Max) (Pf)	t_{rr} (Max) (ns)		
				(V)	(A)	(µA)	(V)					
Power (Continued)	1N5615	1N5615	/429	1.6	3 A	0.5	200	1A	45	150	A248	Fast switching
	1N5617	1N5617					400		35	150		
	1N5619	1N5619					600		25	250		
	1N5621	1N5621					800		20	300		
	1N5623	1N5623					1000		15	500		
	—	1N3891, A, AR	/304	1.5	38 A	10	200	12A (T_C = 100°C)	—	200	DO-203	
		1N3893, A, AR	/304	1.5	38 A	10	400	12A (T_C = 100°C)		200	DO-203	
		1N1202A	/260	1.35	38 (2)	5.0	200	12A (T_C = 150°C)		10 µs	DO-203	Rectifier
	1N5550	1N5550	/420	1.2	9 (2)	1.0	200	3	—	2.0 µs	A1	
	1N5551	1N5551		1.2			400					
	1N5552	1N5552		1.2			600					
	1N5553	1N5553		1.3			800					
	1N5554	1N5554		1.3			1000					
	1N5614	1N5614	/427	1.3	3 (2)	0.5	200	1			A248	
	1N5616	1N5616					400					
	1N5618	1N5618					600					
		1N5620					800					
		1N5622					1000					
	1N5802, US	—	/477	0.975	2.5 (2)	1.0	50	1	25	25	A248	Fast recovery
	1N5804	1N5804		0.975	2.5 (2)	1.0	100	1	25	25	A248	
	1N5806	1N5806		0.975	2.5 (2)	1.0	150	1	25	25	A248	
	1N5807, US	—		0.925	6.0 (2)	5.0	50	3	60	30	A1	
	1N5809	1N5809		0.925	6.0 (2)	5.0	100	3	60	30	A1	
	1N5811	1N5811		0.925	6.0 (2)	5.0	150	3	60	30	A1	
		1N5814	/478	0.95	20 (2)	10.0	100	5	300	35	DO-203	
		1N5816	/478	0.95	20 (2)	10.0	150	5	300	35	DO-203	
Multiple array	—	1N5768	/474	1.0	100mA	0.1	40	300 mA	4	20	TO-86	Common cathode
		1N5770				0.1	40		8	20	TO-86	Common anode
		1N5772				0.1	40		8	20	TO-86	Common anode/cathode
		1N5774				0.1	40		8	20	TO-89	Dual common anode/cathode
		1N6101				25 nA	20		4	5	TO-116	Monolithic

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

(2) Pulsed.

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MIL-S-19500 DIODES, SILICON (Continued)

Description	Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Electrical Characteristics				Pkg Outline
				P_t at T_A = 25°C	V_Z (Nominal) at I_Z (V)	V_Z (Nominal) at I_Z (mA)	V_Z (Reg) or delta BV (V)	
Zener-voltage regulator	1N962B-1 thru 1N973B-1	1N962B-1 thru 1N992B-1	/117	400 mW (T_A = 50°C)	11 - 200	11.5 - 0.65	0.5 - 12	DO-204
	1N754A-1 thru 1N759A-1	1N746A-1 thru 1N759A-1	/127		3.3 - 12	20	1.0 - 0.4	
	—	1N746A thru 1N759A			3.3 - 12	20	1.0 - 0.4	
		1N4370A-1			2.4	20	1.0	
		1N4371A-1			2.7	20	1.0	
		1N4372A-1			3.0	20	1.0	
		1N4614-1 thru 1N4627-1	/435	400 mW	1.8 - 6.2	120 - 61.0	—	
		1N4099-1 thru 1N4135-1	/435	400 mW	6.8 - 100	56 - 3.8		
		1N4565A-1 thru 1N4574A-1	/452 (2)	500 mW	6.4	0.5 - 1.0	5 - 100 mV	
	1N6320 thru 1N6336	—	/533	500 mW (T_L = 75°C)	6.8 - 33	20 - 3.8	1.6 - 0.35	
	1N4464 thru 1N4496	1N6309 thru 1N6336	/533	500 mW (T_L = 75°C)	2.4 - 33	20 - 3.8	1.6 - 0.3	DO-204
		1N3821A thru 1N3828A	/115	1.0 W	3.3 - 6.2	76 - 41	1.0 - 0.3	DO-202
		1N4460 thru 1N4496	/406	1.5 W	6.2 - 200	40 - 1.2	0.35 - 12	A1
	1N4954 thru 1N4992	1N4954 thru 1N4996	/356	2.25 W	6.8 - 360	175 - 3	0.7 - 35	A248
	—	1N2970 thru 1N2977	/124	10 W (T_C = 55°C)	6.8 - 200	370 - 12	0.4 - 9.0	DO-4
		1N2979B thru 1N2980B						
		1N2982B						
		1N2984B thru 1N2986B						
		1N2988B thru 1N2993B						
		1N2995B, 1N2997B						
		1N2999B thru 1N3005B						
		1N3007B thru 1N3009B						
		1N3011B thru 1N3012B						
		1N3014B thru 1N3015B						

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

(2) Reference; temperature-compensated.

MIL-S-19500 DIODES, SILICON (Continued)

Description	Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Electrical Characteristics				Pkg Outline	Remarks
				P_t at T_A = 25°C	V_Z (Nominal) at I_Z (V)	V_Z (Nominal) at I_Z (mA)	V_Z (Reg) or delta BV (mV)		
Zener-voltage regulator (Continued)	—	1N821-1	/159	250 mW	6.2	7.5	96	DO-204	Reference, temperature compensated
		1N823-1					48		
		1N825-1					19		
		1N827-1					9		
		1N829-1					5		
		1N935B-1	/156	500 mW	9.0		184		
		1N937B-1 thru 1N939B-1	/156	500 mW	9.0		37 - 9		
		1N941B	/157	500 mW	11.7		239		
		1N943B	/157	500 mW	11.7		47		
Zener-voltage suppressor		1N5629A thru 1N5665A	/500	1 W	6.8 - 200	10 - 1.0	—	DO-202	Transient voltage - 1500 watts of peak power dissipation at 1.0 ms
		1N5907	/500	1 W	6.4	1.0		DO-202	
		1N5611	/434	3 W	43.7	1.0		DO-203	
Bidirectional voltage suppressor	1N6103 thru 1N6118 1N6103A thru 1N6118A —	1N6103 thru 1N6137	/516	2 W	6.75 - 190	175 - 5.0		A298	500 W
		1N6103A thru 1N6137A		2 W					500 W Peak power
		1N6139 thru 1N6173		3 W					1500 W dissipation
		1N6139A thru 1N6173A		3 W					1500 W at 1.0 ms

(1) Refer to Appendix B for additional testing requirements for JANTX parts.

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MIL-S-19500 DIODES, SILICON (Continued)

Description	Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Electrical Characteristics					Pkg Outline
				P_O (V)	I_P (mA) (Nom)	Z_T (MΩ)	Z_K (MΩ)	V_L (Vdc)	
FET current regulator	—	1N5283 thru 1N5314	/463	100	0.27 - 4.7	14 - 0.235	1.95 - 0.012	1.00 - 2.90	DO-204

Description	Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Electrical Characteristics									Pkg Outline	Remarks
				$V_{(RM)}$ V(pk)	P_T (mW)	$V_{(BR)}$ (10 µAdc) (Vdc)	V_P at I_P (pk)		I_R at V_R		Carrier Lifetime (ps)	C_t (pf)		
							(V)	(mA)	(µA)	(V)				
Schottky barrier	—	1N5711	/444	50	250	70	1.0	15	0.2	50	100	2.0	A1	Hot carrier diode (Schottky)
		1N5712	/445	16	250	20	1.0	35	0.15	16	100	1.2	A1	
		1N6391	/553	45	—	—	0.68	50A (2)	40 mA (2)	45	—	2000	DO-203AA	Pulsed Tests
		1N6392	/554	45			0.68	60A (2)	60 mA (2)	45		3000	DO-203AB	

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

(2) T_C = 125°C, pulsed.

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MIL-S-19500 DIODES, SILICON (Continued)

Description	Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Electrical Characteristics							Pkg Outline	Remarks
				$V_{(RM)}$ V(pk)	V_{FM} (V)	I_{HOX} (mAdc)	I_O (A)	V_{KGM} V (pk)	V_{GT} (Vdc)	t_{on} (µs)		
Thyristor	—	2N2323A 2N2324A 2N2326A 2N2328A	/276	50 100 200 300	2.2	2.0	0.22	6	0.1/0.8	1.0	TO-205	Low-power SCR

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

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MIL-M-38510, MICROCIRCUITS
Digital, Schottky, Low-Power TTL

Description	Generic Type (1)	Function	JAN Part Number (2)		
			M38510	Grade 1	Grade 2
Gates	54LS00	Quad, 2-input positive NAND	/30001	S*X	B*X
	54LS02	NOR, quad 2-input	/30301	S*X(3)	B*X
	54LS03	Quad, 2-input positive NAND (open collector outputs)	/30002	S*X	B*X
	54LS04	Hex inverter	/30003	S*X	B*X
	54LS05	Hex inverter (open collector outputs)	/30004	S*X	B*X
	54LS08	AND, quad 2-input	/31004	S*X(3)	B*X
	54LS09	AND, quad 2-input, open collector outputs	/31005		B*X
	54LS10	Triple, 3-input positive NAND	/30005	S*X	B*X
	54LS11	AND, triple 3-input	/31001	S*X	B*X
	54LS14	Hex inverter Schmitt trigger	/31302	S*X(3)	B*X
	54LS20	Dual, 4-input positive NAND	/30007	S*X(3)	B*X
	54LS21	AND, dual 4-input	/31003	S*X	B*X
	54LS27	NOR, triple 3-input	/30302	S*X	B*X
	54LS30	Single, 8-input positive NAND	/30009	S*X	B*X
	54LS32	OR, quad 2-input	/30501	S*X(3)	B*X
	54LS51	AND-OR-invert, 2 wide 2 input & 2 wide 3 input	/30401	S*X(3)	B*X
	54LS54	AND-OR-invert, 4-wide, 2-3-3-2-input	/30402	S*X(3)	B*X
	54LS86	Exclusive-OR, quad 2-input	/30502	S*X	B*X
	54LS132	Quad 2-input Schmitt trigger, positive NAND	/31303		B*X
	54LS266	Exclusive-NOR, quad 2-input (open collector)	/30303		B*X

(1) Use the JAN M38510 part number for ordering.
(2) The * is for choice of package style. The X is for choice of lead finish. Refer to the QPL for specific choices.
(3) The only remaining QPL supplier for this device will no longer be producing this device type in this grade. These parts are not recommended for new designs. Consult with your supplier for final shipment dates and/or availability information.

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MIL-M-38510, MICROCIRCUITS
Digital, Schottky, Low-Power TTL (Continued)

Description	Generic Type (1)	Function	JAN Part Number (2)		
			M38510	Grade 1	Grade 2
Buffers	54LS26	Quad, 2-input inverter buffer/driver, 15 volt output	/32102	S*X(3)	B*X
	54LS37	Quad, 2-input positive NAND buffer	/30202	S*X(3)	B*X
	54LS38	Quad, 2-input positive NAND buffer (open collector outputs)	/30203	S*X(3)	B*X
	54LS40	Dual, 4-input positive NAND buffer	/30201	S*X(3)	B*X
	54LS125A	Quad, bus buffer (inverting control inputs)	/32301		B*X
	54LS126	Buffer, quad 3-state outputs	/32302		B*X
	54LS240	Octal, inverting buffer (inverting control inputs)	/32401	S*X(3)	B*X
	54LS241	Octal, noninverting buffer (complementary control inputs)	/32402		B*X
	54LS244	Noninverting Octal buffer gates (inverting control inputs)	/32403	S*X	B*X
Drivers	54LS365	Hex, bus driver, gated enable inputs for x-y coincident bus control	/32201	S*X(3)	B*X
	54LS366	Buffer/driver, Hex, inverting, with 3-state outputs	/32202	S*X(3)	B*X
	54LS367	Buffer/driver, Hex, noninverting, with 3-state outputs	/32203	S*X(3)	B*X
	54LS368	Hex, inverter bus driver, 4-line and 2-line enable inputs	/32204	S*X(3)	B*X
Counters	54LS90	Decade counter	/31501	S*X(3)	B*X
	54LS92	Divide-by-twelve counter	/31510		B*X
	54LS93	4-bit binary counter	/31502	S*X(3)	B*X
	54LS160A	Synchronous 4-bit decade (asynchronous clear)	/31503	S*X(3)	B*X
	54LS161A	Synchronous 4-bit binary (asynchronous clear)	/31504	S*X(3)	B*X
	54LS162A	Synchronous 4-bit decade (synchronous clear)	/31511		B*X
	54LS163A	Synchronous 4-bit binary counter (synchronous clear)	/31512	S*X(3)	B*X
	54LS190	Synchronous 4-bit decade counter (with mode control)	/31513	S*X(3)	B*X
	54LS191	Synchronous 4-bit up/down binary (mode control)	/31509	S*X(3)	B*X
	54LS192	Synchronous 4-bit up/down decade counter (dual clock with clear)	/31507		B*X
	54LS193	Synchronous 4-bit up/down binary (dual clock with clear)	/31508	S*X(3)	B*X
	54LS290	Decade counter (BCD or biquinary)	/32003		B*X
	54LS293	4-bit binary counter	/32004		B*X
	54LS390	Dual decade counter with A and B inputs	/32701		B*X
	54LS393	Dual 4-bit binary counter	/32702	S*X(3)	B*X

(1) Use the JAN M38510 part number for ordering.
(2) The * is for choice of package style. The X is for choice of lead finish. Refer to the QPL for specific choices.
(3) The only remaining QPL supplier for this device will no longer be producing this device type in this grade. These parts are not recommended for new designs. Consult with the supplier for final shipment dates and/or availability information.

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MIL-M-38510, MICROCIRCUITS
Digital, Schottky, Fast TTL

Description	Generic Type (1)	Function	JAN Part Number (2)		
			M38510	Grade 1	Grade 2
Gates	54F00	Quad, 2-input positive NAND	/33001	S*X	B*X
	54F02	Quad, 2-input positive NOR	/33301	S*X	B*X
	54F04	Hex inverter	/33002	S*X	B*X
	54F08	AND, quad 2-input	/34001	S*X	B*X
	54F10	Triple, 3-input positive NAND	/33003	S*X	B*X
	54F11	Triple, 3-input positive AND	/34002	S*X	B*X
	54F20	Dual, 4-input positive NAND	/33004	S*X	B*X
	54F32	Quad, 2-input positive OR	/33501	S*X	B*X
	54F64	4-2-3-2 input, positive AND-OR-Invert	/33401	S*X	B*X
	54F86	XOR, quad 2-input	/34501	S*X	B*X
Drivers	54F240	Octal buffers/line drivers with 3-state inverting outputs	/33201	S*X(3)	B*X
	54F241	Octal buffers/line drivers with 3-state outputs	/33202	S*X	B*X
	54F244	Octal buffers/line drivers with 3-state outputs	/33203	S*X	B*X
Counters	54F160A	Synchronous 4-bit decade counter (asynchronous master reset)	/34401	S*X	B*X
	54F161A	Synchronous 4-bit binary counter (asynchronous master reset)	/34301	S*X	B*X
	54F163A	Synchronous presettable binary counter	/34302	S*X	B*X
	54F193	Up/down binary counter with separate up/down clocks	/34304		B*X
Flip-flops	54F74	Dual D-type positive edge-triggered	/34101	S*X	B*X
	54F109	Dual, J-K positive edge-triggered	/34102	S*X(3)	B*X
	54F174	Hex, D-type positive edge-triggered	/34107	S*X	B*X
	54F175	Quad, D-type positive edge-triggered	/34104		B*X
	54F374	Octal, D-type positive edge-triggered with 3-state outputs	/34105		B*X

(1) Use the JAN M38510 part number for ordering.
(2) The * is for choice of package style. The X is for choice of lead finish. Refer to the QPL for specific choices.
(3) The only remaining QPL supplier for this device will no longer be producing this device type in this grade. These parts are not recommended for new designs. Consult with the supplier for final shipment dates and/or availability information.

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MIL-M-38510, MICROCIRCUITS
Digital, Schottky, Fast TTL (Continued)

Description	Generic Type (1)	Function	JAN Part Number (2)		
			M38510	Grade 1	Grade 2
Latch	54F373	Octal transparent latch with active low enable 3-state outputs	/34601	S*X	B*X
	54F533	Octal transparent latch with active low enable 3-state outputs	/34602		B*X
Register	54F398	Quad, 2-port register with complementary outputs	/35001	S*X	B*X
	54F399	Quad, 2-port register	/35002	S*X	B*X
Adder/ Comparator	54F283	4-bit binary full adder with fast carry	/34201	S*X	B*X
	54F521	8-bit identity comparator	/34701	S*X	B*X
Decoders/ MUXs	54F138	1-of-8 decoder, active low outputs with enable inputs	/33701	S*X	B*X
	54F139	Dual 1-of-4 decoder, active low outputs with enable inputs	/33702	S*X	B*X
	54F151A	8-input multiplexer	/33901	S*X	B*X
	54F153	Dual 4-input multiplexer	/33902	S*X	B*X
	54F157A	Quad 2-input multiplexer	/33903	S*X	B*X
	54F158A	Quad 2-input multiplexer with inverted outputs	/33904	S*X	B*X
	54F251A	8-input multiplexer with 3-state outputs	/33905	S*X	B*X
	54F253	Dual 4-input multiplexer with 3-state outputs	/33908	S*X	B*X
	54F257A	Quad 2-input multiplexer with 3-state outputs	/33906	S*X	B*X
	54F258A	Quad 2-input multiplexer with 3-state inverted outputs	/33907	S*X	B*X
	54F352	Dual 4-input multiplexer with inverted outputs	/33909		B*X
	54F353	Octal transparent latch with active low enable, 3-state outputs	/33910		B*X
Combinational	54F280	9-bit odd/even parity generator/checker	/34901	S*X	B*X
Transceiver	54F245	Octal noninverting bus transceiver w/ 3-state outputs	/34803	S*X	B*X

(1) Use the JAN M38510 part number for ordering.
(2) The * is for choice of package style. The X is for choice of lead finish. Refer to the QPL for specific choices.

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MIL-M-38510, MICROCIRCUITS
Digital, Interface

Description	Generic Type (1)	Function	JAN Part Number (2)		
			M38510	Grade 1	Grade 2
Digital to Analog Converters	DAC-08	8-bit D/A converters, 0.19% linearity	/11301	S*X	B*X
	DAC-08A	8-bit D/A converters, 0.10% linearity	/11302	S*X	B*X
	562	12 bit digital to analog converter, external reference	/12101		B*X
	565	12 bit digital to analog converter, internal reference, high speed	/12103		B*X
	7521	12 bit D/A converter with 10 bit end-point linearity	/12703		B*X
	561	10 bit binary D/A converter, internal reference	/13301		B*X
Analog to Digital Converters	574AT	12 bit, analog to digital converter with microprocessor interface	/14002		B*X
	574AU	12 bit, analog to digital converter with microprocessor interface	/14001		B*X

(1) Use the JAN M38510 part number for ordering.
(2) The * is for choice of package style. The X is for choice of lead finish. Refer to the QPL for specific choices.

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MIL-M-38510, MICROCIRCUITS
Bipolar, Memories

Description	Generic Type (1)	Function	JAN Part Number (2)		
			M38510	Grade 1	Grade 2
RAM	93L422	Static, 256 x 4, Low power, 3-state output	/23112		B*X
PROM	82S23	32 x 8 with open collector outputs	/20701		B*X(3)
	82S115	512 x 8 with 3-state outputs	/20803		B*X(3)
	82S123A	32 x 8 with 3-state output	/20704	S*X(3)	B*X(3)
	82S126	256 x 4 with open collector	/20301		B*X(3)
	82S129	256 x 4 with active pull-up and 3-state output	/20302	S*X(3)	B*X(3)
	82S129A	256 x 4 with active pull-up and 3-state output	/20304		B*X(3)
	82S130	512 x 4 with open collector	/20401		B*X(3)
	82S131	512 x 4 with active pull-up and 3-state output	/20402		B*X(3)
	82S131A	512 x 4 with active pull-up and 3-state output	/20404		B*X(3)
	82S137	1K x 4 with 3-state output	/20602		B*X(3)
	82S137A	1K x 4 with 3-state outputs	/20604	S*X(3)	B*X(3)
	82S141	512 x 8 with active pull-up and 3-state output	/20802		B*X(3)
	82S181	1K x 8 with active pull-up and 3-state output	/20904	S*X(3)	B*X
	82S181A	1K x 8 with active pull-up and 3-state output	/20909		B*X(3)
	82S191	2K x 8 with active pull-up and 3-state output	/21002	S*X(3)	B*X
	82S185	2K x 4 with active pull-up and 3-state output	/20902		B*X
	82S185A	2K x 4 with active pull-up and 3-state output	/20910		B*X(3)
	82S191A	2K x 8 with active pull-up and 3-state output	/21004	S*X(3)	B*X(3)

(1) Use the JAN M38510 part number for ordering.
(2) The * is for choice of package style. The X is for choice of lead finish. Refer to the QPL for specific choices.
(3) The only remaining QPL supplier for this device will no longer be producing this device type in this grade. These parts are not recommended for new designs. Consult with the supplier for final shipment dates and/or availability information.

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GSFC S-311-P-18, THERMISTORS

Part Number Explanation:

311P18	-XX	X	XXX
GSFC Control Specification	Dash Number corresponds to the zero-power resistance of the thermistor @ +25°C	Lead Material: S = 32 AG, type C per MIL-STD-1276 T = 28 AG, type ET per MIL-W-16878 N = 32 AG, type N-2 per MIL-STD-1276 E = insulated lead, TFE, 32 AG per MIL-I-22129; bare lead, style S; FEP tubing, M23053/11-105C	Lead Length (cm) (4)

Part Number	Control Specification	Temperature Coefficient	Seal	Zero Power Resistance (at +25°C) (Ohms)	Tolerance Limits (0 to +70°C) (±%)	Operating and Storage Temperature Limits(°C)	Thermal Time Constant (sec)Max	Dissipation Constant (mW/°C) Min	FRL Grades 1 and 2
311P18-01XXXX	GSFC S-311-P-18	Negative	Nonhermetic	2252	1.02	-55 to +90	10.0 (1)	1.0 (2)	(3)
311P18-02XXXX				2252	0.51	-55 to +70			
311P18-03XXXX				3000	1.02	-55 to +90			
311P18-04XXXX				3000	0.56	-55 to +70			
311P18-05XXXX				5000	1.02	-55 to +90			
311P18-06XXXX				5000	0.51	-55 to +70			
311P18-07XXXX				10000	0.93	-55 to +90			
311P18-08XXXX				10000	0.56	-55 to +70			
311P18-09XXXX				30000	1.00	-55 to +90			
311P18-10XXXX				30000	0.50	-55 to +70			

(1) For a thermistor suspended in still air, the thermal constant is 10.0 seconds maximum, except lead style "E", 25 seconds maximum.
(2) For thermistors suspended in still air only (does not apply for lead style "E").
(3) Parts may be used in Grades 1 and 2 applications.
(4) Standard lead lengths are from 7.6 cm to 100 cm. The lead length in centimeters (cm) is specified by three characters. For lengths from 7.6 cm to 9.9 cm the letter "R" is used as the decimal point (e.g., 7R6 (7.6)). For lengths 10 cm to 99 cm the letter "R" is used as the last character (e.g., 76R (76)). For 100 cm the characters are 101, where the last character signifies the number of zeros following the first two characters.

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SECTION 13: SUMMARY OF STANDARD TRANSISTORS

Page	Control Specification	Description	Type Designation	
			Grade 1	Grade 2 (1)
13.2	MIL-S-19500	Low-power, NPN	JANS	JANTXV
13.3		Low-power, PNP		
13.3		Matched pair, NPN, PNP		
13.4		High Power, NPN, PNP		
13.5		Silicon RF, NPN, PNP		
13.6		FETs, N-channel, P-channel		
13.7		Choppers		
13.8		Optocoupler		

Refer to MIL-HDBK-978, Vol. 2, for construction and application information.

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

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MIL-S-19500, TRANSISTORS
Silicon, Low Power (Less Than 2W)

Part Number Explanation:

JAN — Military Qualified

AAA — Quality Level: S = Grade 1; TXV = Grade 2

X (1) — Radiation Hardness Assurance (RHA)

RHA Designator	Total Dose Radiation Tolerance RAD (Si)	Level Neutron Fluence (n/cm)
M	3×10^3	2×10^{12}
D	1×10^4	2×10^{12}
R	1×10^5	1×10^{12}
H	1×10^6	1×10^{12}

YN — Component Designator: Y = number one less than the number of active element terminations.

BBBB — Identification Number

I — Suffix Letter: A, B, C, etc. (excluding L, M, R, S) - indicates a modified version which is substitutable for the non-suffixed device.

- M - indicates matching of specified parameters of separate devices.
- R - indicates reverse polarity packaging of the non-suffixed device.
- L or S - indicates that the terminal leads are longer or shorter, respectively, than those of the non-suffixed device.

Grade 1 JANS	Grade 2 JANTXV (2)	Specification MIL-S-19500	Pkg Outline	Maximum Ratings: P_T at T_A = +25°C	Maximum Ratings: BV_{CBO} (V)	Maximum Ratings: BV_{CEO} (V)	Maximum Ratings: I_C (A)	Electrical Characteristics: h_{FE} at I_C Min/Max	h_{FE} at I_C	V_{CE}(SAT) at I_C (V)	V_{CE}(SAT) at I_C	I_{CBO} (Max) at V_{CB} (Max)	Switching Time t_{on} (ns)	Switching Time t_{off} (ns)	Remarks
Low Power NPN															
2N2219A.AL	2N2219A.AL	/251	TO-205	800 mW	75	50	0.8	100/300	150 mA	1.0	500 mA	10 nA	35	300	Comp to 2N2905A
2N2222A	2N2222A	/255	TO-206	500 mW	75	50	0.8	100/300	150 mA	1.0	500 mA	10 nA	35	300	Comp to 2N2907A
2N2369A	2N2369A	/317	TO-206	360 mW	40	15	0.2	40/120	10 mA	0.2	10 mA	0.2 µA	12	18	F_T = 500 MHz min
2N2484	2N2484	/376	TO-206	360 mW	60	60	0.05	225/800	10 mA	0.3	1.0 mA	5 nA	—	—	—
2N3019	2N3019,S	/391	TO-205	800 mW	140	80	1.0	100/300	150 mA	0.2	150 mA	10(I_{CES}) nA	—	—	$t_{on} + t_{off}$ = 30 ns
2N3418,S	2N3418,S	/393	TO-205	1.0 W	85	60	3.0	20/60	1.0 A	0.25	1.0 A	—	300	1300	—
2N3419,S	2N3419,S	/393	TO-205	1.0 W	125	80	3.0	20/60	1.0 A	0.25	1.0 A	—	300	1300	—
2N3420,S	2N3420,S	/393	TO-205	1.0 W	85	60	3.0	40/120	1.0 A	0.25	1.0 A	—	300	1300	—
2N3735,L	—	/395	TO-205	1.0 W	75	40	1.5	40/140	0.5 A	0.5	50 mA	100 nA @ 20V	43	115	—

(1) Included only for JANTXV and JANS product assurance level devices.

(2) Refer to Appendix B for additional testing requirements for JANTXV parts.

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MIL-S-19500, TRANSISTORS
Silicon, Low Power (Less Than 2W) (Continued)

Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Pkg Outline	Maximum Ratings				Electrical Characteristics							Remarks
								h_{FE} at I_C		V_{CE}(SAT) at I_C		I_{CBO} (Max) at V_{CB} (Max)	Switching Time		
				P_T at T_A = +25°C	BV_{CBO} (V)	BV_{CEO} (V)	I_C (A)	Min/Max		(V)			t_{on} (ns)	t_{off} (ns)	
Low Power PNP															
2N3421,S	2N3421,S	/393	TO-205	1.0 W	125	80	3.0	40/120	1.0 A	0.25	1.0 A		300	1300	—
2N3501L	2N3501L	/366	TO-205	1.0 W	150	150	0.3	100/300	150 mA	0.4	150 mA	50 nA @75V	115	1150	—
2N3700	2N3700	/391	TO-206	500 mW	140	80	1.0	100/300	150 mA	0.2	150 mA	10 (I_{CES}) nA			$t_{on} + t_{off}$ = 30 ns
2N4150,S	2N4150,S	/394	TO-205	1.5 W	100	70	10	40/120	5 A	0.6	5 A	100 nA	550	2000	—
2N5237,S	—	/394	TO-205	1.5 W	150	120	10	40/120	5 A	0.6	5A	100 nA	550	2000	
2N5238,S	—	/394	TO-205	1.5 W	200	170	10	40/120	5 A	0.6	5 A	100 nA	550	2000	—
2N5666	2N5666	/455	TO-205	1.2 W	250	200	5	40/120	1 A	0.4	3 A	1.0 mA	250	1500	—
2N5667	2N5667	/455	TO-205	1.2 W	400	300	5	25/75	1 A	0.4	3 A	1.0 mA	250	2000	—
2N2605	2N2605	/354	TO-206	400 mW	70	60	0.03	100/300	0.010 mA	0.3	10 mA	10 nA			low-noise amp
2N2905A,AL	2N2905A,AL	/290	TO-205	600 mW	60	60	0.6	100/300	150 mA	0.4	150 mA	10 nA	45	300	Comp to 2N2219A
2N2907A	2N2907A	/291	TO-206	400 mW	60	60	0.6	100/300	150 mA	0.4	150 mA	10 nA	45	300	Comp to 2N2222A
	2N3251A	/323	TO-206	360 mW	60	60	0.2	100/300	10 mA	0.25	10 mA	20 nA @ 40V	70	250	—
	2N3468,L	/348	TO-205	1.0 W	50	50	1.0	25/75	500 mA	0.6	500 mA	100 nA @ 30V	40	90	—
2N3634,L	—	/357	TO-205	1.0 W	140	140	1.0	50/100	50 mA	0.6	50 mA	100 nA @ 100V	200	650	
2N3635,L	—	/357	TO-205	1.0 W	140	140	1.0	100/300	50 mA	0.6	50 mA	100 nA @ 100V	200	650	
2N3636,L	—	/357	TO-205	1.0 W	175	175	1.0	50/150	50 mA	0.6	50 mA	100 nA @ 100V	200	650	
2N3637,L	2N3637,L	/357	TO-205	1.0 W	175	175	1.0	100/300	50 mA	0.6	50 mA	100 nA @ 100V	200	650	—
	2N3763,L	/396	TO-205	1.0 W	60	60	1.5	40/140	500 mA	0.5	500 mA	100 nA @ 30V	43	115	—
2N3764	—	/396	TO-205	0.5 W	60	60	1.5	40/140	0.5 A	0.5	50 mA	100 nA @ 20V	43	115	
	2N3765	/396	TO-206	500 mW	60	60	1.5	40/140	500 mA	0.5	500 mA	100 nA @ 30V	43	115	—
2N3867,S	2N3867,S	/350	TO-205	1.0 W	40	40	3.0	40/200	1.5 A	0.75	1.5 A	—	100	600	—
2N3868,S	2N3868,S	/350	TO-205	1.0 W	60	60	3.0	30/150	1.5 A	0.75	1.5 A	—	100	600	—
	2N5416,S	/485	TO-205	750 mW	350	300	1.0	30/120	50 mA	2.0	50 mA	0.05 mA	1000	10000	—
NPN Matched Pair															
2N2060	2N2060,L	/270	TO-78	600(both) mW	100	60	0.5	40/120	1 mA	0.3	50 mA	2 nA	—	—	h_{FE} matched±10%,
2N2919	2N2919,L	/355	TO-99	500(both) mW	70	60	0.03	60/240	10 µA	0.3	1.0 mA	2 nA	—	—	V_{BE} matched±5mV
2N2920	2N2920,L	/355	TO-99	500(both) mW	70	60	0.03	175/600	10 µA	0.3	1.0 mA	2 nA	—	—	—
PNP Matched Pair															
2N3810	2N3810,L	/336	TO-78	600(both) mW	60	60	0.05	150/450	1 mA	0.25	1.0 mA	10 nA	—	—	h_{FE} matched±10%,
2N3811	2N3811,L	/336	TO-78	600(both) mW	60	60	0.05	300/900	1 mA	0.25	1.0 mA	10 nA	—	—	V_{BE} matched±5 mV

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

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MIL-S-19500, TRANSISTORS
Silicon, High-Power (Greater Than 2 W)

Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Pkg Outline	Maximum Ratings				Electrical Characteristics							Remarks
								h_{FE} at I_C		$V_{CE(SAT)}$ at I_C		I_{CBO} (Max) at V_{CB} (Max) (µA)	Switching Time		
				P_T at T_C = +100°C (W)	BV_{CBO} (V)	BV_{CEO} (V)	I_C (A)	Min/Max	(A)	(V)	(A)		t_{on} (µs)	t_{off} (µs)	
High-Power NPN															
	2N3716	/408	TO-204	85.7	100	80	10	50/150	1.0	1.0	5	1.0 mA (I_{CES}) @ 70 V	1.5	2.4	Comp to 2N3792
2N3749	2N3749	/315	MT53	30	110	80	5	40/120	1.0	0.25	1	0.2	0.36	2.0	Similar to 2N2880. Isolated collector
2N2880	—	/315	MT53	30	110	80	5	40/120	1.0	0.25	1	0.2	0.3	1.7	—
2N3996	2N3996	/374	TO-111	30	100	80	5	40/120	1.0	2.0	5	0.2(I_{CES})	0.3	1.5	—
2N3997	—	/374	TO-111	30	100	80	5	80/240	1.0	2.0	5	0.2(I_{CES})	0.3	2.0	—
2N5038	2N5038	/439	TO-204	80	150	90	20	50/200	2.0	1.0	12	1.0	0.5	2.0	—
2N5039	—	/439	TO-204	80	125	75	20	30/150	2.0	1.0	1.0	1.0	0.5	2.0	—
2N5664	2N5664	/455	TO-213	30	250	200	5	40/120	1.0	0.4	3	1000	0.25	1.5	—
2N5665	2N5665	/455	TO-213	30	400	300	5	25/75	1.0	0.4	3	1000	0.25	2.0	—
	2N5672	/488	TO-204	80	150	120	30	20/100	15.0	0.75	15	25 mA	0.5	1.5	—
	2N6308	/498	TO-204	62.5	700	350	8	12/60	3.0	1.5	3	500(I_{CEX})	0.60	3.0	—
High-Power PNP															
2N3740	—	/441	TO-213	14	60	60	4	30/120	0.25	0.6	1	0.1	0.4	1.0	—
2N3741	2N3741	/441	TO-213	14	80	80	4	30/120	0.25	0.6	1	0.1	0.4	1.0	—
	2N3792	/379	TO-204	85.7	80	80	10	50/150	1.0	1.0	5	0.5 mA (I_{CES}) @ 70 V	1.5	2.4	Comp to 2N3716
—	2N4399	/433	TO-204	115	60	60	30	25/—	15.0	1.0	15	1000	1.2	2.5	—
	2N5745	/433	TO-204	115	80	80	20	25/—	10.0	1.5	15	1000	1.2	2.5	—
	2N6546	/525	TO-204	100	—	300	15	12/60	5.0	1.5	10	—	1.0	4.7	Pulsed $V_{CE(SAT)}$
	2N6547	/525	TO-204	100	—	400	15	12/60	5.0	1.5	10	—	1.0	4.7	Pulsed $V_{CE(SAT)}$

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

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MIL-S-19500, TRANSISTORS
Silicon, RF Devices, NPN, PNP

Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Pkg Outline	Maximum Ratings		Electrical Characteristics							Remarks
				P_T T_C = +25°C (W)	BV_{CBO} (Volts)	h_{FE} (Min/Max)	$\lvert h_{fe} \rvert$ at f Min/Max	(MHz)	P_{OUT} (Min/Max) (Watts)	@ P_{IN} (Watts)	f (MHz)	n (%)	
NPN													
2N918	2N918	/301	TO-206	0.3	30	20/200	6/18	100	G_{PE} = 15db min	—	200	—	—
2N2857	2N2857	/343	TO-206	0.3	30	30/150	10/21	100	G_{PE} = 12.5/21 db	—	450	—	—
	2N3375	/341	TO-60	11.6	65	15/150	3.5 min	100	7.5/14	1	100	65	—
	2N3553	/341	TO-205	7.0	65	15/150	3.5 min	100	2.5/5	0.25	175	50	—
	2N3866	/398	TO-205	1.0 @ T_A = 25°C	60	15/200	2.5/8	200	1.0/2.0	0.15	400	45	—
2N3866A	—	/398	TO-205	1.0 @ T_A = 25°C	60	25/200	2.5/8	200	1.0/2.0	0.15	400	45	—
2N5109	—	/453	TO-39	1.0 W	40	40/120	6/11	200	G_{PE} = 11dB	10dBm	200	—	—
PNP													
	2N4957	/426	TO-206	0.2 @ T_A = 25°C	30	30/165	12/36	100	G_{PE} = 17/25 db	—	450	—	—

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

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Supersedes Page 13.6 of MIL-STD-975L

MIL-S-19500, TRANSISTORS
Silicon, Field Effect

Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Pkg Outline	Maximum Ratings				Electrical Characteristics						Remarks
				P_T at T_A=+25°C (mW)	V_{DG} (V)	V_{DS} (V)	I_G (mA)	$V_{GS(OFF)}$ (Min/Max) (V)	I_{DSS} (V_{GS}=0) (Min/Max) (mA)	$r_{ds(ON)}$ Max (Ohm)	y_{fs} (µmho)	C_{iss} Max (pF)	C_{rss} Max (pF)	
N-Channel														
2N3821	2N3821	/375	TO-206	300	50	50	10	-4 max	0.5/2.5	—	1500/4500	6.0	3.0	$\|y_{fs}\|$ @ 1 kHz
2N3822	2N3822	/375	TO-206	300	50	50	10	-6 max	2/10	—	3000/6500	6.0	3.0	$\|y_{fs}\|$ @ 1 kHz
	2N3823	/375	TO-206	300	30	30	10	-8 max	4/20	—	3500/6500	6.0	2.0	$\|y_{fs}\|$ @ 1 kHz
	2N4416A	/428	TO-206	300	35	35	10	-2.5/-6.0	5/15	—	4500/7500	4.0	0.8	$\|y_{fs}\|$ @ 1 kHz
	2N4856	/385	TO-206	360	40	40	50	-4/-10	50/175	25	—	18	8.0	Analog Sw/Chopper
	2N4857	/385	TO-206	360	40	40	50	-2/-6	20/100	40	—	18	8.0	Analog Sw/Chopper
	2N4858	/385	TO-206	360	40	40	50	-0.8/-4	8/80	60	—	18	8.0	Analog Sw/Chopper
N-Channel Radiation Hardened														
2N7261	—	/601	TO-205	800	100	100	—	—	—/-0.050	0.180	—	—	—	Radiation Level R
2N7262	—	/601	TO-205	800	200	200	—	—	—/-0.050	0.400	—	—	—	Radiation Level H
2N7268	—	/603	TO-254	4000	100	100	—	—	—/-0.050	0.070	—	—	—	Radiation Level R
2N7269	—	/603	TO-254	4000	200	200	—	—	—/-0.050	0.115	—	—	—	Radiation Level H
P-Channel														
	2N5114	/476	TO-206	500	30	30	50	5/10	-30/-90	75	—	25	7.0	Analog Sw/Chopper
	2N5115	/476	TO-206	500	30	30	50	3/6	-15/-60	100	—	25	7.0	Analog Sw/Chopper
	2N5116	/476	TO-206	500	30	30	50	1/4	-5/-25	175	—	27	7.0	Analog Sw/Chopper
	2N6804	/562	TO-204	4000	-100	-100	—	—	—/-0.025	0.3	—	800	200	—
	2N6806	/562	TO-204	4000	-200	-200	—	—	—/-0.025	0.8	—	800	90	—
	2N6845	/563	TO-205	800	-100	-100	—	—	—/-0.025	0.6	—	400	100	—
	2N6847	/563	TO-205	800	-200	-200	—	—	—/-0.025	1.50	—	400	45	—

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

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MIL-S-19500, TRANSISTORS
Silicon, Chopper

Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Pkg Outline	Maximum Ratings				Electrical Characteristics					Remarks
								h_{FE} at I_C		$V_{EC(ofs)}$ at $I_E = 0$		$r_{ec(on)}$	
				P_T at $T_A = +25°C$ (mW)	BV_{CBO} (V)	BV_{CEO} (V)	I_C (mA)	Min/Max	(mA)	Min/Max (mV)	I_B (mA)	Min/Max (Ohms)	
2N2432A	2N2432A	/313	TO-206	300	45	45	100	80/400	1	—/0.7	1	—/15	NPN
	2N2945A	/382	TO-206	400	-25	-20	-100	70/—	-1	—/-1.0	-1	—/6	PNP

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

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MIL-S-19500, TRANSISTORS
Optocoupler

Grade 1 JANS	Grade 2 JANTXV (1)	Specification MIL-S-19500	Pkg Outline	Maximum Ratings: P_T at T_A=+25°C (W)	Maximum Ratings: V_{CEO} (Max) (V)	Maximum Ratings: V_R (Max) (V)	Maximum Ratings: I_C Max (mAdc)	Maximum Ratings: I_F Max (mAdc)	Electrical Characteristics: $I_{C(ON)}$ V_{CE} = 5.0 V Min/Max (mAdc)	I_F (mA)	Electrical Characteristics: $V_{CE(SAT)}$ @ I_C & I_F Min/Max (V)	(mA)	(mA)	Electrical Characteristics: $I_{C(OFF)}$ @ V_{CE} I_C & I_B = 0 Min/Max (nA)	(V)	Remarks
—	4N23	/486	PH13	0.3	35	2	50	40	6.0/—	10	—/0.3	5	20	—/100	20	The designation A denotes isolated phototransistor.
	4N23A	/486			35				6.0/—	10		5	20			
	4N24	/486			35				10.0/—	10		10	20			
	4N24A	/486			35				10.0/—	10		10	20			
	4N47	/548			40				0.5/—	1.0		0.5	2.0			These devices are recommended for new design.
	4N48	/548							1.0/5.0			1.0	2.0			
	4N49	/548							2.0/10.0			2.0	2.0			

(1) Refer to Appendix B for additional testing requirements for JANTXV parts.

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Microcircuit Information

Part Numbering System for NASA Detail Part Specifications:

Part Number							
Specification Number				Part Number Suffix			
NA62	X	XX	XXX	XX	S	X	X
Stock Class Designator NA62 is the designator for microcircuits based on the Federal Stock Class.	Total Dose Radiation Hardness Designator 3 krad M 5 krad N 10 krad D 20 krad E 30 krad F 50 krad G 100 krad R 200 krad S 300 krad T 500 krad U 1000 krad H	Drawing Number Year Last two digits of the year the drawing number was assigned.	Drawing Number Sequential drawing number within the fiscal year.	Type Number The type number within the drawing.	Class Designator Class S- Equivalent	Package Designator Package designator per MIL-M-38510.	Lead Finish Designator Lead finish per MIL-M-38510.

Package Type Abbreviations:

DIP = Dual In-line Package; FP = Flatpack Package; LCC = Leadless Chip Carrier; PGA = Pin Grid Array.

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INTERFACE HYBRID MICROCIRCUITS (46 parts)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
MIL-STD-1553 (1553) Driver-Receiver, Single Ch.	5962-86049 DESC	ILC#	02TX	BUS-63106II	FP	24	H	NA	-15 V
1553 Driver-Receiver, Single Ch.	5962-86049 DESC	ILC#	02ZX	BUS-63105II	DIP	24	H	NA	-15 V
1553 Driver-Receiver, Single Ch.	5962-86049 DESC	ARX#	03YX	ARX-2402	DIP	24	H	NA	+/-15 V
1553 Driver-Receiver, Single Ch.	5962-86049 DESC	NHI#	06YX	NHI-1509	DIP	24	H	NA	+15 V
1553 Dual Redundant Remote Terminal Unit (RTU)	5962-87535 DESC	ILC#	01XX	BUS-65112	Hybrid	78	H	NA	+/-15 V
1553 Dual Redundant RTU	5962-87535 DESC	ILC#	01YX	BUS-65117	FP	82	H	NA	+/-15 V
1553 Driver-Receiver, Dual Ch.	5962-87579 DESC	ILC#	02XX	BUS-63125II	DIP	36	H	NA	-15 V
1553 Driver-Receiver, Dual Ch.	5962-87579 DESC	ILC#	02YX	BUS-63126II	FP	36	H	NA	-15 V
1553 Driver-Receiver, Dual Ch.	5962-87579 DESC	MRL#	08XX	MR63125MPR	DIP	36	H	NA	-15 V
12-Bit Data Acquisition System (DAS)	5962-88514 DESC	Sipex#	01XX	HS9403B-8	Quad	62	H	NA	+/-15 V
12-Bit DAS	5962-88514 DESC	Sipex#	02XX	HS9403B-16	Quad	62	H	NA	+/-15 V
1553 Bus Controller (BC), RTU, and Monitor Unit (MT) Dual Redundant	5962-88585 DESC	ILC#	01XX	BUS-65600	DIP	78	H	NA	+5 V
1553 Bus-to-Microprocessor Interface Unit	5962-88586 DESC	ILC#	01XX	BUS-66300II-883B	DIP	78	H	NA	+7 V
1553 Bus-to-Microprocessor Interface Unit	5962-88586 DESC	ILC#	01YX	BUS-66301II-883B	FP	82	H	NA	+7 V

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INTERFACE HYBRID MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	01XC	BUS-61553	DIP	78	H	NA	+5, -15 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	01XX	BUS-61553	DIP	78	H	NA	+5, -15 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	01XX	BUS-61563	DIP	78	H	NA	+5, -15 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	01YA	BUS-61563	FP	82	H	NA	+5, -15 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	01YC	BUS-61563	FP	82	H	NA	+5, -15 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	01YX	BUS-61563	FP	82	H	NA	+5, -15 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	02XA	BUS-61554	DIP	78	H	NA	+5, -12 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	02XC	BUS-61554	DIP	78	H	NA	+5, -12 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	02XX	BUS-61554	DIP	78	H	NA	+5, -12 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	02YC	BUS-61564	FP	82	H	NA	+5, -12 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	03XC	BUS-61555	DIP	78	H	NA	+5 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	03YC	BUS-61565	FP	82	H	NA	+5 V
1553 BC/RTU/MT Multiplexed Terminal	5962-88692 DESC	ILC#	04XC	BUS-61556	DIP	78	H	NA	+5 V
1553 Driver-Receiver, Dual Ch. Receiver Idle	5962-89447 DESC	ILC#	04HXX	BUS-63135II	DIP	36	H	NA	-15 V

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INTERFACE HYBRID MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
1553 Driver-Receiver, Dual Ch. Receiver Idle	5962-89447 DESC	ILC#	04HYX	BUS-63136II	FP	36	H	NA	-15 V
1553 Transceiver, Dual Ch.	5962-89522 DESC	ARX#	02XX	ARX-2453	DIP	36	H	NA	+5 V
Dual 8 Channel MUX w/Precision Sample	5962-89565 DESC	Sipex#	01XX	HS362SB	DIP	32	H	NA	+/-15 V
Dual Redundant RTU	5962-89798 DESC	ILC#	01XX	BUS-65142	Hybrid	78	H	NA	+5 V
Dual Redundant RTU	5962-89798 DESC	ILC#	01YX	BUS-65144	FP	82	H	NA	+5 V
Dual Redundant RTU	5962-89798 DESC	ILC#	02XX	BUS-65143	Hybrid	78	H	NA	+5 V
Dual Redundant RTU	5962-89798 DESC	ILC#	02YX	BUS-65145	FP	82	H	NA	+5 V
1553 Driver-Receiver, Dual Ch.	5962-89826 DESC	ILC#	01XX	BUS-63127II	DIP	36	H	NA	-12 V
1553 Driver-Receiver, Dual Ch.	5962-89826 DESC	ILC#	01YX	BUS-63128II	FP	36	H	NA	-12 V
1553 Data Terminal Bit Processor	5962-90636 DESC	MED#	01HYX	CT1820F	FP	60	H	NA	+5 V
1553 Data Terminal Bit Processor	5962-90636 DESC	MED#	02HYX	CT1820-2F	FP	60	H	NA	+5 V
1553 Data Terminal Bit Processor	5962-90636 DESC	ARX#	03HXX	ARX2410	Plug-In	56	H	NA	+5 V
1553 Data Terminal Bit Processor	5962-90636 DESC	ARX#	04HXX	ARX3410	Plug-In	56	H	NA	+5 V
1553 MUX Bus RT	5962-91687 DESC	NHI#	01HXX	NHI-1553RT	DIP	66	H	NA	+5 V

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INTERFACE HYBRID MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
1553 MUX Bus RT	5962-91687 DESC	NHI#	01HYX	NHI-1553RTFP	FP	66	H	NA	+5 V
1553 MUX Bus RT	5962-91687 DESC	NHI#	02HXX	NHI-1554RT	DIP	66	H	NA	+5 V
1553 MUX Bus RTU	5962-91687 DESC	NHI#	02HYX	NHI-1554RTFP	FP	66	H	NA	+5 V
16 Channel MUX/DeMUX	PT40872 JPL	HAR	None	HS 1840	FP	28	S-Equi	200 Krad	+15 V

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INTERFACE MONOLITHIC MICROCIRCUITS (1 part)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
1553B BC/RT/M	NA62R91006 JPL	UNT@	01SKC	UT1553B BCRTMWSR	FP	84	S-Equi	100 Krad	+7 V

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ANALOG-TO-DIGITAL HYBRID MICROCIRCUITS (29 parts)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
12-Bit A/D Converter (ADC)	5962-88508 DESC	Datel#	01XX	ADC-HX	DIP	32	H	NA	+/-15 V
12-Bit ADC	5962-88508 DESC	Datel#	02XX	ADC-HZ	DIP	32	H	NA	+/-15 V
12-Bit ADC	5962-88508 DESC	MN#	02XX	MNADC87H/B	DIP	32	H	NA	+/-15 V
12-Bit ADC Track and Hold	5962-88542 DESC	ILC#	01XX	ADC-00300-112	DIP	40	H	NA	+/-15 V
12-Bit ADC	5962-88658 DESC	ADI#	01XX	AD578SD	DIP	32	H	NA	+/-15 V
12-Bit ADC	5962-88658 DESC	ADI#	02XX	AD578TD	DIP	32	H	NA	+/-15 V
12-Bit ADC	5962-88658 DESC	ADI#	03XX	AD578ZSD	DIP	32	H	NA	+/-12 V
12-Bit ADC	5962-88658 DESC	ADI#	04XX	AD578ZTD	DIP	32	H	NA	+/-15 V
12-Bit ADC	5962-89569 DESC	MN#	01HXX	MN5295H/B	DIP	32	H	NA	+/-15 V
12-Bit ADC	5962-89583 DESC	MN#	01YX	MN5200H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89583 DESC	MN#	02YX	MN5203H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89583 DESC	MN#	03YX	MN5201H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89583 DESC	MN#	04YX	MN5204H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89583 DESC	MN#	05YX	MN5202H/B	DIP	24	H	NA	+/-15 V

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ANALOG-TO-DIGITAL HYBRID MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
12-Bit ADC	5962-89583 DESC	MN#	06YX	MN5205H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89583 DESC	MN#	07YX	MN5206H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89584 DESC	MN#	01YX	MN5210H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89584 DESC	MN#	02YX	MN5213H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89584 DESC	MN#	03YX	MN5211H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89584 DESC	MN#	04YX	MN5214H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89584 DESC	MN#	05YX	MN5212H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89584 DESC	MN#	06YX	MN5215H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89584 DESC	MN#	07YX	MN5216H/B	DIP	24	H	NA	+/-15 V
12-Bit ADC	5962-89595 DESC	MN#	01XX	MN5245H/B	DIP	40	H	NA	+/-15 V
12-Bit ADC	5962-89595 DESC	MN#	02XX	MN5245AH/B	DIP	40	H	NA	+/-15 V
12-Bit ADC	5962-89595 DESC	MN#	03XX	MN5246H/B	DIP	40	H	NA	+/-15 V
12-Bit ADC	5962-89595 DESC	MN#	04XX	MN5246AH/B	DIP	40	H	NA	+/-15 V
16-Bit High Speed ADC	5962-90795 DESC	Sipex#	01HXX	HS9576S/B	DIP	32	H	NA	+/-15 V

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ANALOG-TO-DIGITAL HYBRID MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
16-Bit High Speed ADC	5962-90795 DESC	Sipex#	02HXX	HS9576T/B	DIP	32	H	NA	+/-15 V

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ANALOG-TO-DIGITAL MONOLITHIC MICROCIRCUITS (1 part)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
16-Bit ADC	S-311-P-704 GSFC	Crystal	None	CS5016	DIP	40	S-Equi	10 Krad	+/-6 V

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DIGITAL-TO-ANALOG HYBRID MICROCIRCUITS (20 parts)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Bits
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
12-Bit D-to-A Range Programmable Voltage	83002 DESC	ADI#	01JX	DAC87	DIP	24	H	NA	12
12-Bit DAC Bipolar Linear Quad	5962-88509 DESC	ADI#	01XX	AD390SD	DIP	28	H	NA	12
12-Bit DAC Bipolar Linear Quad	5962-88509 DESC	ADI#	02XX	AD390TD	DIP	28	H	NA	12
12-Bit DAC Bipolar Quad	5962-88510 DESC	ADI#	01XX	AD394SD	DIP	28	H	NA	12
12-Bit DAC Bipolar Quad	5962-88510 DESC	ADI#	02XX	AD394TD	DIP	28	H	NA	12
12-Bit DAC Bipolar Quad	5962-88510 DESC	ADI#	03XX	AD395SD	DIP	28	H	NA	12
12-Bit DAC Bipolar Quad	5962-88510 DESC	ADI#	04XX	AD395TD	DIP	28	H	NA	12
14-Bit DAC De-glitched	5962-88517 DESC	ILC#	01XA	DAC02310-112	DIP	32	H	NA	14
14-Bit DAC De-glitched	5962-88517 DESC	ILC#	01XC	DAC02310-112	DIP	32	H	NA	14
14-Bit DAC De-glitched	5962-88517 DESC	ILC#	01YA	DAC02311-112	FP	32	H	NA	14
14-Bit DAC De-glitched	5962-88517 DESC	ILC#	01YC	DAC02311-112	FP	32	H	NA	14
14-Bit DAC De-glitched	5962-88517 DESC	ILC#	02XA	DAC02310-113	DIP	32	H	NA	14
14-Bit DAC De-glitched	5962-88517 DESC	ILC#	02XC	DAC02310-113	DIP	32	H	NA	14
14-Bit DAC De-glitched	5962-88517 DESC	ILC#	02YA	DAC02311-113	FP	32	H	NA	14

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DIGITAL-TO-ANALOG HYBRID MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Bits
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
14-Bit DAC De-glitched	5962-88517 DESC	ILC#	02YC	DAC02311-113	FP	32	H	NA	14
12-Bit DAC Binary Input Code	5962-89528 DESC	Datel#	01XX	DAC HKB	DIP	24	H	NA	12
12-Bit DAC 2s Complement	5962-89528 DESC	Datel#	02XX	DAC HKB-2	DIP	24	H	NA	12
16-Bit DAC	5962-89531 DESC	Datel#	01HXX	DAC HPB	DIP	24	H	NA	16
16-Bit DAC	5962-89531 DESC	Datel#	02HXX	DAC HPB-1	DIP	24	H	NA	16
16-Bit Track and Hold Amplifier	5962-90630 DESC	Sipex#	01HXX	SP9760B	DIP	24	H	NA	16

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DIGITAL-TO-SYNCHRO HYBRID MICROCIRCUITS (4 parts)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Bits
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
16-Bit Digital-to-Synchro and Digital-to-Resolver Converter	5962-89986 DESC	ILC#	01HXX	DSC-11520-112	DIP	36	H	NA	16
16-Bit Digital-to-Synchro and Digital-to-Resolver Converter	5962-89986 DESC	ILC#	02HXX	DSC-11520-113	DIP	36	H	NA	16
16-Bit Digital-to-Synchro and Digital-to-Resolver Converter	5962-89986 DESC	ILC#	03HXX	DSC-11520-114	DIP	36	H	NA	16
16-Bit Digital-to-Synchro and Digital-to-Resolver Converter	5962-89986 DESC	ILC#	04HXX	DSC-11520-115	DIP	36	H	NA	16

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PROCESSORS/CONTROLLERS and PERIPHERALS MONOLITHIC MICROCIRCUITS (44 parts)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Bits
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
16-Bit Parallel EDAC	54HSC/T630 MED	MED#	None	54HSC/T630	DIP	28	S-Equi	1 Mrad	16
16-Bit Parallel EDAC	54HSC/T630 MED	MED#	None	54HSC/T630	FP	28	S-Equi	1 Mrad	16
32-Bit 20 MHz Microprocessor	5962-87668 DESC	INT**	03QXX	80386-20	PGA	132	Q	2.5 Krad	32
32-Bit 20 MHz Microprocessor	5962-87668 DESC	INT**	03QYX	80386	LCC	164	Q	2.5 Krad	32
32-Bit 20 MHz Microprocessor	5962-87668 DESC	INT**	04QXX	80386	PGA	132	Q	2.5 Krad	32
32-Bit 20 MHz Microprocessor	5962-87668 DESC	INT**	04QYX	80386	LCC	164	Q	2.5 Krad	32
80-Bit 20 MHz Numeric Processor Extension	5962-89534 DESC	INT**	02QYX	MQ80387Q-20/Q	FP	68	Q	NA	80
80-Bit 25 MHz Numeric Processor Extension	5962-89534 DESC	INT**	03QYX	MQ80387Q-25/Q	FP	68	Q	NA	80
16-Bit Digital Signal Processor	5962-90704 DESC	AT&T**	01QXX	WE-DSP-16	PGA	133	Q	1 Mrad	16
16-Bit 1750 Microprocessor, Fixed Point	5962-92026 DESC	IBM**	01QZX	FXPC-Q49F	FP	220	Q	2 Mrad	16
16-Bit 1750 Microprocessor, Fixed Point	5962-92026 DESC	IBM**	01VZX	FXPC-V49F	FP	220	V	2 Mrad	16
16-Bit 1750 Microprocessor, Floating Point	5962-92103 DESC	IBM**	01QZX	MFLPC-Q47F	LCC	220	Q	2 Mrad	16
16-Bit 1750 Microprocessor, Floating Point	5962-92103 DESC	IBM**	01VZX	MFLPC-V47F	LCC	220	V	2 Mrad	16
1750 Microprocessor Address Processor 1	5962-92104 DESC	IBM**	01QZX	176A685	FP	220	Q	2 Mrad	NA

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PROCESSORS/CONTROLLERS and PERIPHERALS MONOLITHIC MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Bits
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
1750 Microprocessor Address Processor 1	5962-92104 DESC	IBM**	01VZX	176A685	FP	220	V	2 Mrad	NA
1750 Microprocessor Address Processor 2	5962-92106 DESC	IBM**	01QZX	165A984	FP	220	Q	2 Mrad	NA
1750 Microprocessor Address Processor 2	5962-92106 DESC	IBM**	01VZX	165A984	FP	220	V	2 Mrad	NA
16-Bit Parallel EDAC	ACS630MS HAR	HAR	None	ACS630MS	DIP	28	S-Equi	1 Mrad	16
16-Bit Parallel EDAC	ACS630MS HAR	HAR	None	ACS630MS	FP	28	S-Equi	1 Mrad	16
16-Bit Parallel EDAC	ACTS630MS HAR	HAR	None	ACTS630MS	DIP	28	S-Equi	1 Mrad	16
16-Bit Parallel EDAC	ACTS630MS HAR	HAR	None	ACTS630MS	FP	28	S-Equi	1 Mrad	16
8-Bit Microprocessor	CS515800 JPL	HAR	1	80C85	FP	42	S-Equi	100 Krad	8
Peripheral to 80C85 (2K-Bit SRAM with I/O and Timer)	CS515801 JPL	HAR	1	81C55	FP	42	S-Equi	100 Krad	2K
Programmable DMA Controller	HS-82C37ARH HAR	HAR	None	HS-82C37ARHR	FP	42	S-Equi	100 Krad	16
5 MHz Priority Interrupt Controller	NA62-91003 JPL	HAR	01S3A	MR82C59A	LCC	28	S-Equi	NA	24
5 MHz Priority Interrupt Controller	NA62-91003 JPL	HAR	01SJC	MD82C59A	DIP	28	S-Equi	NA	24
16-Bit Microprocessor	NA62R91001 JPL	HAR	01SQC	HS1-80C86RH	DIP	40	S-Equi	100 Krad	16
16-Bit Microprocessor	NA62R91001 JPL	HAR	01SXC	HS9-80C86RH	FP	42	S-Equi	100 Krad	16

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PROCESSORS/CONTROLLERS and PERIPHERALS MONOLITHIC MICROCIRCUITS (contd)

SPECIFICATION		*PART*					*Quality*	*Radiation*	*Bits*
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads	Level	(TID)	
5 MHz Programmable Interval Timer	NA62R91002 JPL	HAR	01SJC	HS1-82C54RH	DIP	24	S-Equi	100 Krad	16
5 MHz Programmable Interval Timer	NA62R91002 JPL	HAR	01SKC	HS9-82C54RH	FP	24	S-Equi	100 Krad	16
5 MHz Static Clock Controller/Generator	NA62R91004 JPL	HAR	01SJC	HS1-82C85RH	DIP	24	S-Equi	100 Krad	24
5 MHz Static Clock Controller/Generator	NA62R91004 JPL	HAR	01SKC	HS9-82C85RH	FP	24	S-Equi	100 Krad	24
Priority Interrupt Controller	S-311-P-708 GSFC	HAR	1	82C59A-1	FP (shielded)	28	S-Equi	100 Krad	24
16-Bit 8 MHz Microcontroller	S-311-P-721 GSFC	HAR	001	RTX2010RH-Q	FP	84	S-Equi	1 Mrad	16
16-Bit 8 MHz Microcontroller	S-311-P-721 GSFC	HAR	002	RTX2010RH-Q	PGA	84	S-Equi	1 Mrad	16
16-Bit Microprocessor	SSQ22662 SSF	HAR	01SQX	HS-80C86RH	DIP	40	S-Equi	100 Krad	16
Programmable Interval Timer	SSQ22663 SSF	HAR	01SJX	HS-82C54RH	DIP	24	S-Equi	100 Krad	16
5 MHz Programmable Interrupt Controller	SSQ22665 SSF	HAR	01SYX	MD82C59A-5/B	DIP	28	S-Equi	NA	24
16-Bit Microprocessor	SSQ22667 SSF	INT**	001	SQ80C186-12	Quad FP	68	S-Equi	NA	16
32-Bit Microprocessor	SSQ22668 SSF	INT**	001	SQ80386-25	Quad FP	164	S-Equi	2.5 Krad	32
80-Bit Numeric Processor	SSQ22669 SSF	INT**	001	SQ80387-25	FP	68	S-Equi	NA	80
Multi-Bus II Interface Controller	SSQ22670 SSF	INT**	001	SQ82389	Quad FP	164	S-Equi	NA	32

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PROCESSORS/CONTROLLERS and PERIPHERALS MONOLITHIC MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Bits
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
1553 Bus Controller, Remote Terminal	SSQ22673-001 SSF	UNT@	001	UT1553BBCRT WS	FP	84	S-Equi	NA	16
32-Bit DMAC	SSQ22692 SSF	INT**	001	82380-25	LCC	164	S-Equi	NA	32

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RESOLVER/SYNCHRO-TO-DIGITAL CONVERTERS HYBRID MICROCIRCUITS (20 parts)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Accuracy (Arc Min)
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
14-Bit 11.8 V Resolver or Synchro-to-Digital Converter	5962-87687 DESC	ILC#	01XX	HSDC-8920A	DIP	36	H	NA	+/-2.6
14-Bit 11.8 V Resolver or Synchro-to-Digital Converter	5962-87687 DESC	ILC#	02XX	HSDC-8920	DIP	36	H	NA	+/-4.0
90.0 V Synchro-to-Digital Converter	5962-89498 DESC	ILC#	01XX	SDC14561-114	DIP	36	H	NA	+/-2.0
90.0 V Synchro-to-Digital Converter	5962-89498 DESC	ILC#	02XX	SDC14561-115	DIP	36	H	NA	+/-1.0
16-Bit 11.8 V Synchro-to-Digital Converter	5962-89499 DESC	ILC#	03XX	SDC14560-605	DIP	36	H	NA	+/-2.0
16-Bit 11.8 V Synchro-to-Digital Converter	5962-89908 DESC	ILC#	03XX	SDC14565-616	DIP	36	H	NA	+/-2.0
16-Bit 1.0 V Direct Resolver-to-Digital	5962-90551 DESC	ILC#	01XX	SDC14567-111	DIP	36	H	NA	+/-6.0
16-Bit 1.0 V Direct Resolver-to-Digital	5962-90551 DESC	ILC#	02XX	SDC14567-112	DIP	36	H	NA	+/-4.0
16-Bit 1.0 V Direct Resolver-to-Digital	5962-90551 DESC	ILC#	03XX	SDC14567-114	DIP	36	H	NA	+/-2.0
16-Bit 1.0 V Direct Resolver-to-Digital	5962-90551 DESC	ILC#	04XX	SDC14567-115	DIP	36	H	NA	+/-1.0
1.0 V Direct Resolver-to-Digital Converter	5962-90551 DESC	ILC#	05XX	SDC14569-111	DIP	36	H	NA	+/-6.0
1.0 V Direct Resolver-to-Digital Converter	5962-90551 DESC	ILC#	06XX	SDC14569-112	DIP	36	H	NA	+/-4.0
1.0 V Direct Resolver-to-Digital Converter	5962-90551 DESC	ILC#	07XX	SDC14569-114	DIP	36	H	NA	+/-2.0
1.0 V Direct Resolver-to-Digital Converter	5962-90551 DESC	ILC#	08XX	SDC14569-115	DIP	36	H	NA	+/-1.0

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RESOLVER/SYNCHRO-TO-DIGITAL CONVERTERS HYBRID MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Accuracy (Arc Min)
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
14-Bit Resolver or Synchro-to-Digital	5962-90554 DESC	ILC#	01XX	HSDC-8915A	DIP	36	H	NA	+/-2.6
14-Bit Resolver or Synchro-to-Digital	5962-90554 DESC	ILC#	02XX	HSDC-8915	DIP	36	H	NA	+/-4.0
14-Bit 11.8 V Synchro-to-Digital Converter	5962-90707 DESC	ILC#	01HXX	SDC14500-635	DIP	36	H	NA	+/-2.6
14-Bit 11.8 V Synchro-to-Digital Converter	5962-90707 DESC	ILC#	02HXX	SDC14500-636	DIP	36	H	NA	+/-4.0
14-Bit 90.0 V Synchro-to-Digital Converter	5962-90707 DESC	ILC#	03HXX	SDC14502-605	DIP	36	H	NA	+/-2.6
14-Bit 90.0 V Synchro-to-Digital Converter	5962-90707 DESC	ILC#	04HXX	SDC14502-606	DIP	36	H	NA	+/-4.0

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MEMORIES MONOLITHIC MICROCIRCUITS (42 parts)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Access Time
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
8Kx8 SRAM (Has data retention.)	5962-38294 DESC	HON**	43Q9X	HC6364/1XQHCC	FP	36	Q	2 Mrad	55 ns
8Kx8 SRAM (Has data retention.)	5962-38294 DESC	HON**	43V9X	HC6364/1XVHCC	FP	36	V	2 Mrad	55 ns
8Kx8 SRAM (No data retention.)	5962-38294 DESC	IBM**	44Q8X	146A894	FP	36	Q	2 Mrad	55 ns
8Kx8 SRAM (No data retention.)	5962-38294 DESC	IBM**	44V8X	146A894	FP	36	V	2 Mrad	55 ns
32Kx8 SRAM	5962-92153 DESC	IBM**	01QXX	IBM2568C-Q60X	FP	36	Q	1 Mrad	60 ns
32Kx8 SRAM	5962-92153 DESC	IBM**	01QYX	IBM2568C-Q60Y	FP	40	Q	1 Mrad	60 ns
32Kx8 SRAM	5962-92153 DESC	IBM**	01VXX	IBM2568C-V60X	FP	36	V	1 Mrad	60 ns
32Kx8 SRAM	5962-92153 DESC	IBM**	01VYX	IBM2568C-V60Y	FP	40	V	1 Mrad	60 ns
UT7156 32Kx8 SRAM	5962-92153 DESC	UNT@	11QMX	UT7156	FP	40	Q	1 Mrad	55 ns
UT7156 32Kx8 SRAM	5962-92153 DESC	UNT@	11QMX	UT7156	FP	40	Q	100 Krad	55 ns
UT7156 32Kx8 SRAM	5962-92153 DESC	UNT@	12QTX	UT7156	FP	40	Q	1 Mrad	55 ns
UT7156 32Kx8 SRAM	5962-92153 DESC	UNT@	12QTX	UT7156	FP	40	Q	100 Krad	55 ns
64Kx1 SRAM	5962-92154 DESC	IBM**	01QXX	IBM6401C-Q55X	FP	36	Q	1 Mrad	55 ns
64Kx1 SRAM	5962-92154 DESC	IBM**	01QYX	IBM6401C-Q55G	LCC	32	Q	1 Mrad	55 ns

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MEMORIES MONOLITHIC MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Access Time
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
64Kx1 SRAM	5962-92154 DESC	IBM**	01VXX	IBM6401C-V55X	FP	36	V	1 Mrad	55 ns
64Kx1 SRAM	5962-92154 DESC	IBM**	01VYX	IBM6401C-V55G	LCC	32	V	1 Mrad	55 ns
64Kx1 SRAM	5962-92154 DESC	IBM**	02QXX	IBM6401T-Q55X	FP	36	Q	1 Mrad	55 ns
64Kx1 SRAM	5962-92154 DESC	IBM**	02QYX	IBM6401T-Q55X	LCC	32	Q	1 Mrad	55 ns
64Kx1 SRAM	5962-92154 DESC	IBM**	02VXX	IBM6401T-V55X	FP	36	V	1 Mrad	55 ns
64Kx1 SRAM	5962-92154 DESC	IBM**	02VYX	IBM6401T-V55X	LCC	32	V	1 Mrad	55 ns
256Kx1 SRAM	5962-93128 DESC	IBM**	01QXX	IBM2561C-Q60X	FP	36	Q	1 Mrad	60 ns
256Kx1 SRAM	5962-93128 DESC	IBM**	01VXX	IBM2561C-V60X	FP	36	V	1 Mrad	60 ns
256Kx1 SRAM	5962-93128 DESC	IBM**	02QXX	IBM2561T-Q60X	FP	36	Q	1 Mrad	60 ns
256Kx1 SRAM	5962-93128 DESC	IBM**	02VXX	IBM2561T-V60X	FP	36	V	1 Mrad	60 ns
256Kx1 SRAM	5962-93128 DESC	IBM**	03QXX	IBM2561C-Q40X	FP	36	Q	1 Mrad	40 ns
256Kx1 SRAM	5962-93128 DESC	IBM**	03VXX	IBM2561C-V40X	FP	36	V	1 Mrad	40 ns
256Kx1 SRAM	5962-93128 DESC	IBM**	04QXX	IBM2561T-Q40X	FP	36	Q	1 Mrad	40 ns
256Kx1 SRAM	5962-93128 DESC	IBM**	04VXX	IBM2561T-V40X	FP	36	V	1 Mrad	40 ns

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MEMORIES MONOLITHIC MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Access Time
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
32Kx8 SRAM	5962H92153 DESC	IBM**	02QXX	IBM2568T-Q60X	FP	36	Q	1 Mrad	60 ns
32Kx8 SRAM	5962H92153 DESC	IBM**	02VXX	IBM2568T-V60X	FP	36	V	1 Mrad	60 ns
32Kx8 SRAM	5962H92153 DESC	HON**	05QUX	HC6856/1WQHZC	FP	36	Q	1 Mrad	40 ns
32Kx8 SRAM	5962H92153 DESC	HON**	05VUX	HC6856/1WVHZC	FP	36	V	1 Mrad	40 ns
64Kx1 SRAM	HS-65643RHR HAR	HAR	None	HS-65643RHR	FP	28	S-Equi	300 Krad	35 ns
8Kx8 SRAM	HS-65647RH-Q HAR	HAR	None	HS-65647RH-Q	DIP	28	S-Equi	300 Krad	50 ns
8Kx8 SRAM	HS-65647RH-Q HAR	HAR	None	HS-65647RH-Q	FP	28	S-Equi	300 Krad	35 ns
32Kx8 SRAM	HS-65758RH-Q HAR	HAR	None	HS-65758RH-Q	DIP	28	S-Equi	100 Krad	35 ns
32Kx8 SRAM	HS-65758RH-Q HAR	HAR	None	HS-65758RH-Q	FP	28	S-Equi	100 Krad	35 ns
32Kx8 SRAM	HS-65759RH-Q HAR	HAR	None	HS-65759RH-Q	FP	36	S-Equi	100 Krad	35 ns
2Kx8 PROM	NA62R91005 JPL	HAR	01SJC	6617RH	DIP	24	S-Equi	100 Krad	50 ns
2Kx8 PROM	NA62R91005 JPL	HAR	01SKC	6617RH	FP	24	S-Equi	100 Krad	50 ns
512x9 FIFO	S-311-P-705 GSFC	MED#	1	7001RH	DIP	28	S-Equi	100 Krad	90 ns
512x9 FIFO	S-311-P-705 GSFC	MED#	2	7001RH	FP	28	S-Equi	100 Krad	90 ns

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ASIC MONOLITHIC MICROCIRCUITS (22 parts)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
9000 Gate Programmable Logic Array	5962-89823 DESC	AT&T**	01QXX	ATT3090-50R175MQ	PGA	175	Q	NA	+5 V
9000 Gate Programmable Array	5962-89823 DESC	AT&T**	01QZC	ATT3090-50N164MQ	PGA	164	Q	NA	+5 V
9000 Gate Programmable Logic Array	5962-89823 DESC	AT&T**	02QXX	ATT3090-70R175MQ	PGA	175	Q	NA	+5 V
9000 Gate Programmable Array	5962-89823 DESC	AT&T**	02QZC	ATT3090-70N164MQ	PGA	164	Q	NA	+5 V
ASIC, Common Microcontroller	5962-92A02 DESC	AT&T**	01QYA	1056A2	LCC	256	Q	NA	+7 V
ASIC, Control Bus Interface/Functional (CBI/FBIT)	5962-92A03 DESC	AT&T**	01QYA	1056B2	LCC	256	Q	NA	+7 V
ASIC, Parallel Input/Output Port	5962-92A04 DESC	AT&T**	01QYA	1056C	LCC	256	Q	NA	+7 V
ASIC, Operand Memory	5962-92A06 DESC	AT&T**	01QYA	1056E	LCC	256	Q	NA	+7 V
ASIC, Dual Multiplexer Operand Memory	5962-92A07 DESC	AT&T**	01QYA	1056F	LCC	256	Q	NA	+7 V
ASIC, Cache Bus Controller	5962-92A08 DESC	AT&T**	01QYX	1056G	LCC	256	Q	NA	+7 V
ASIC, PROTEUS Digital Channel	5962-92A09 DESC	AT&T**	01QYA	1056H2	LCC	256	Q	NA	+7 V
ASIC, ISC Arithmetic Unit (IAU)	5962-92A10 DESC	AT&T**	01QYX	1056J	LCC	256	Q	NA	+7 V
ASIC, Input Controller	5962-92A12 DESC	AT&T**	01QYX	1056L	LCC	256	Q	NA	+7 V
ASIC, Switch	5962-92A17 DESC	AT&T**	01QYA	1057A3	LCC	132	Q	NA	+7 V

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ASIC MONOLITHIC MICROCIRCUITS (contd)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
ASIC, Sequencer	5962-92A18 DESC	AT&T**	01QYA	1057C	LCC	132	Q	NA	+7 V
ASIC, Error Detection and Correction	5962-92A19 DESC	AT&T**	01QYA	1057D	LCC	132	Q	NA	+7 V
ASIC, ISC Digital-to-Analog Controller	5962-92A21 DESC	AT&T**	01QYA	1057F2	LCC	132	Q	NA	+7 V
ASIC, Address Controller	5962-92A22 DESC	AT&T**	01QYX	1057G	LCC	132	Q	NA	+7 V
Serial Communication Controller	5962H92A01 DESC	HON**	01QYX	1HK84	FP	84	Q	1 Mrad	+5 V
Serial Communication Controller	5962H92A01 DESC	HON**	01VYX	1HK84	FP	84	V	1 Mrad	+5 V
Trace Data Processor, Semicustom Gate	5962R93A01 DESC	HON**	01QYX	1HL08	SCC	256	Q	100 Krad	+6 V
ASIC, Gate Array Input Output Unit (IOU)	ST12186 JPL	UNT@	UTD160R	12186-E0UTD160FR	FP	84	S-Equi	100 Krad	+7 V

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PALS MONOLITHIC MICROCIRCUITS (1 part)

SPECIFICATION		PART					Quality Level	Radiation (TID)	Supply Voltage
Title	No./Source	Vendor	Type Number	Generic Type	Package	Leads			
Programmable Array Logic	95916 LML	TIX#	001	PAL22VP1025MJT	DIP	24	S-Equi	15 Krad	+5 V

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IDENTIFICATION CODE LIST*

CODE	NAME
ADI#	Analog Devices, Inc.
ARX#	Aeroflex Laboratories, Inc.
AT&T**	American Telephone & Telegraph
Crystal	Crystal Semiconductor
Datel#	Datel, Inc.
DESC	Defense Electronics Supply Center
GSFC	Goddard Space Flight Center
HAR	Harris Semiconductor
HON**	Honeywell, Inc.
IBM**	International Business Machines
ILC#	ILC Data Device Corporation
INT**	Intel Corporation
JPL	Jet Propulsion Laboratory
LML	Lockheed Monitored Line
MED#	Marconi Circuit Technology (GEC Plessey)
MN#	MicroNetworks Company
MRL#	Micro-Rel Div. of Medtronic Incorporated
NHI#	National Hybrid, Inc.
Sipex#	Sipex Corporation
SSF	Space Station Freedom
STEL	Stanford Telecom
TIX#	Texas Instruments, Inc.
UNT@	United Technologies Microelectronics Center, Inc.

* These codes appear in the Advanced Microcircuit Parts Listing table under the column headers "No./Source." and "Vendor."
** These manufacturers are QML Certified and Qualified for QML-38535.
These manufacturers are QML Certified and Qualified for QML-38534.
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